

EventPulse

AI-Powered Event-Driven Traffic Forecasting

Predict congestion and optimize manpower deployment using AI — before incidents escalate into gridlock.



Made with GAMMA

Traffic Challenges in Urban Events

Urban events — from VIP movements to water logging — trigger congestion that manual planning cannot address in time.



VIP Movements

Unplanned road closures



Public Gatherings

Protests, rallies, events



Construction & Potholes

Recurring infrastructure disruptions



Water Logging

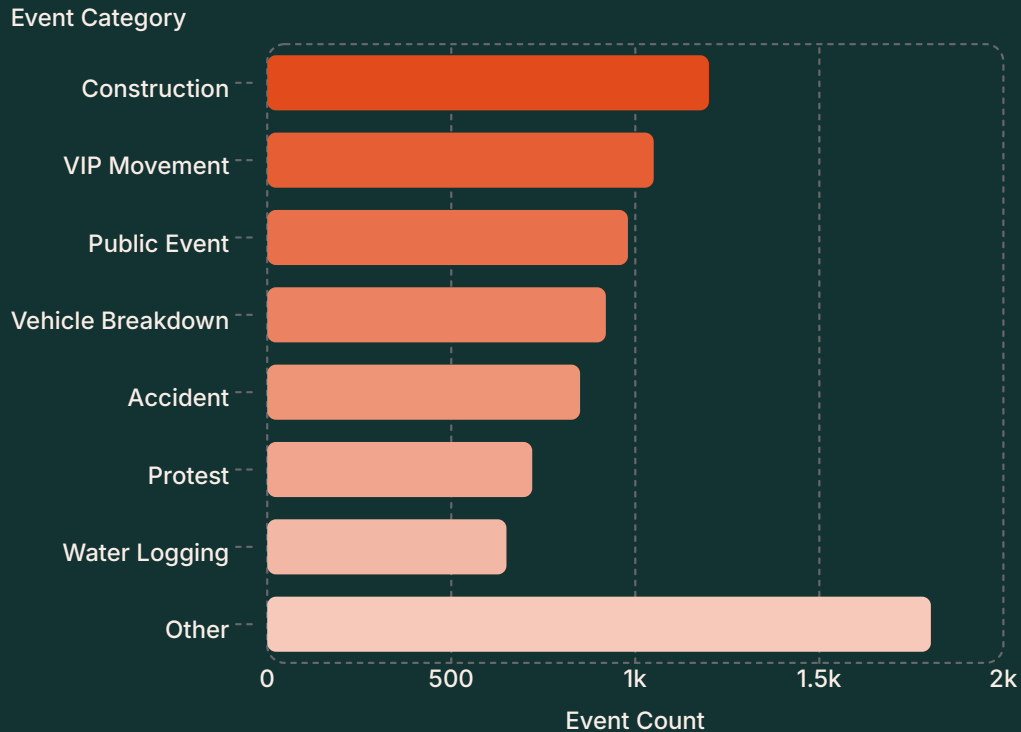
Monsoon-induced blockages



Current Gaps: Manual planning · Delayed response · No impact estimation · Poor diversion strategies



ASTraM Bengaluru Event Dataset

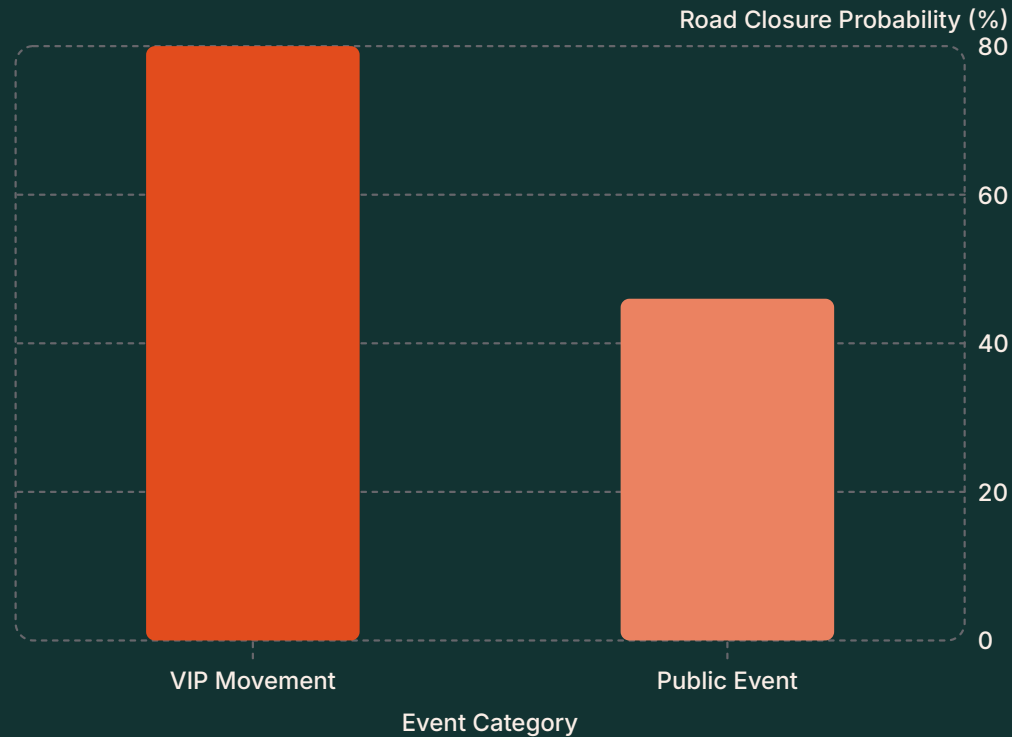


Dataset at a Glance

ASTraM's Bengaluru dataset powers EventPulse's predictive engine — covering real-world urban disruption scenarios across the city.

- **8,173** total recorded events
- **16** distinct event categories
- Spans construction, accidents, VIP movements, protests, and more
- Enables granular, category-specific risk modeling

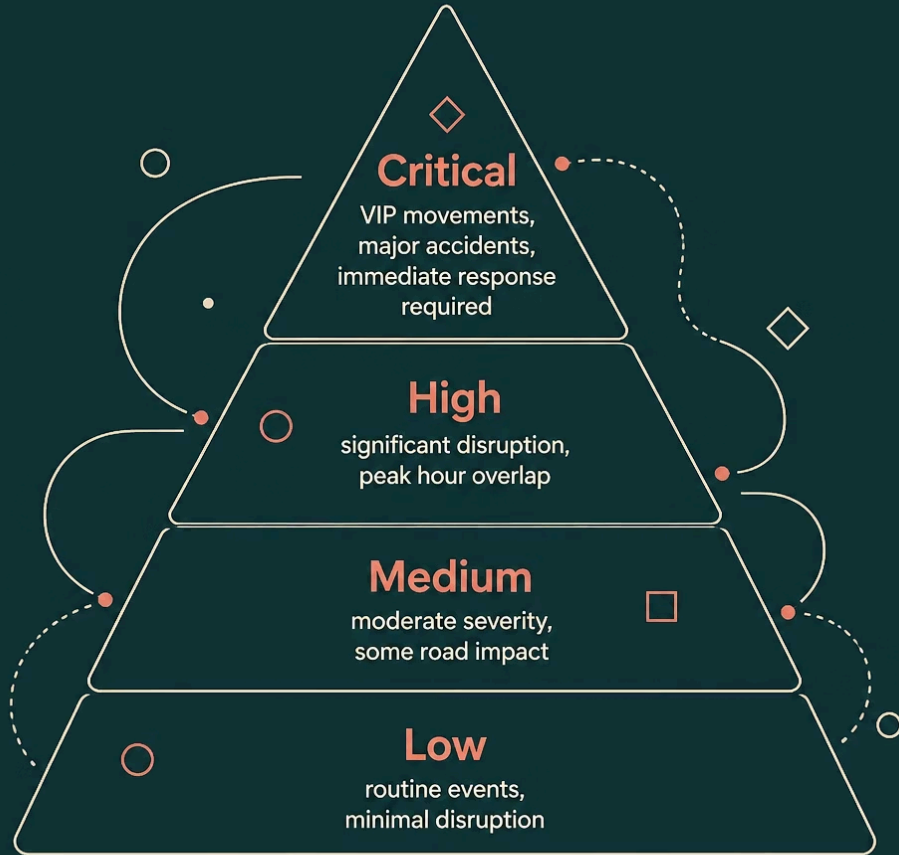
Understanding Event Patterns



Key EDA Findings

- **Peak hours** identified across all event categories
- **VIP movements** require road closure in **80%** of cases
- **Public events** require closure in **46%** of cases
- **Potholes peak on Saturdays** – a recurring weekend pattern
- Weekday vs. weekend event distributions reveal distinct behavioral trends

AI-Based Risk Assessment



Risk Scoring Methodology

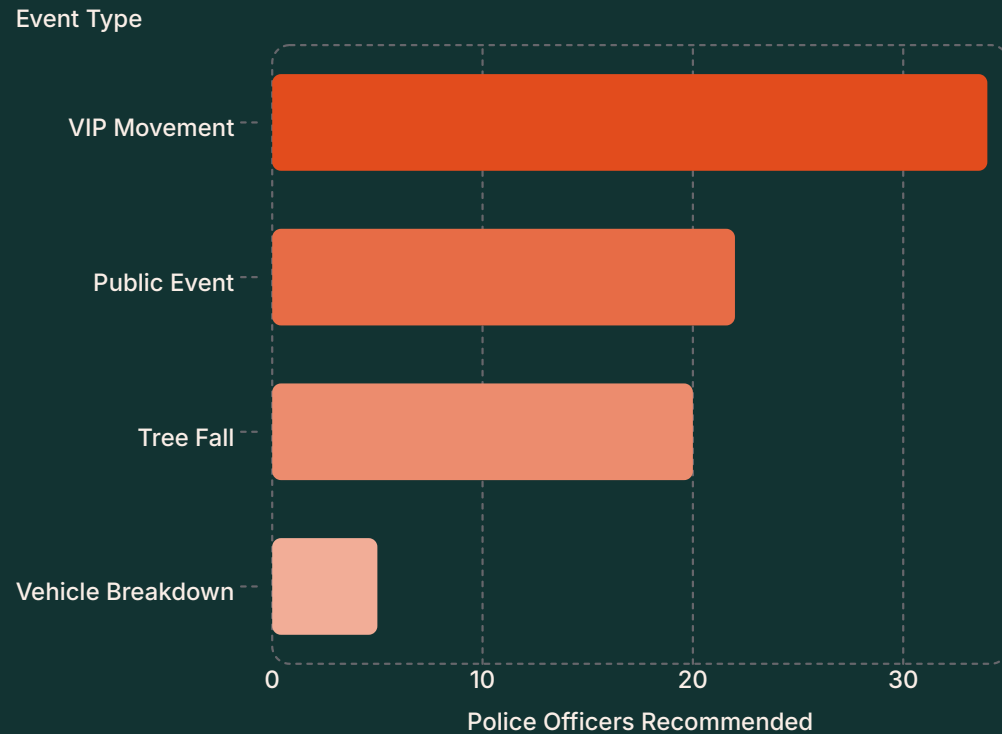
Each event is scored across four dimensions to produce a composite risk classification:

- **Severity score** – based on event type and historical impact
- **Peak hour indicator** – time-of-day sensitivity weighting
- **Road closure requirement** – binary closure probability flag
- **High-risk day analysis** – weekend and holiday weighting

□ Average severity score across all events: ≈ 2.27

Intelligent Resource Recommendation

EventPulse translates risk scores into precise, actionable resource allocations — reducing over- and under-deployment.

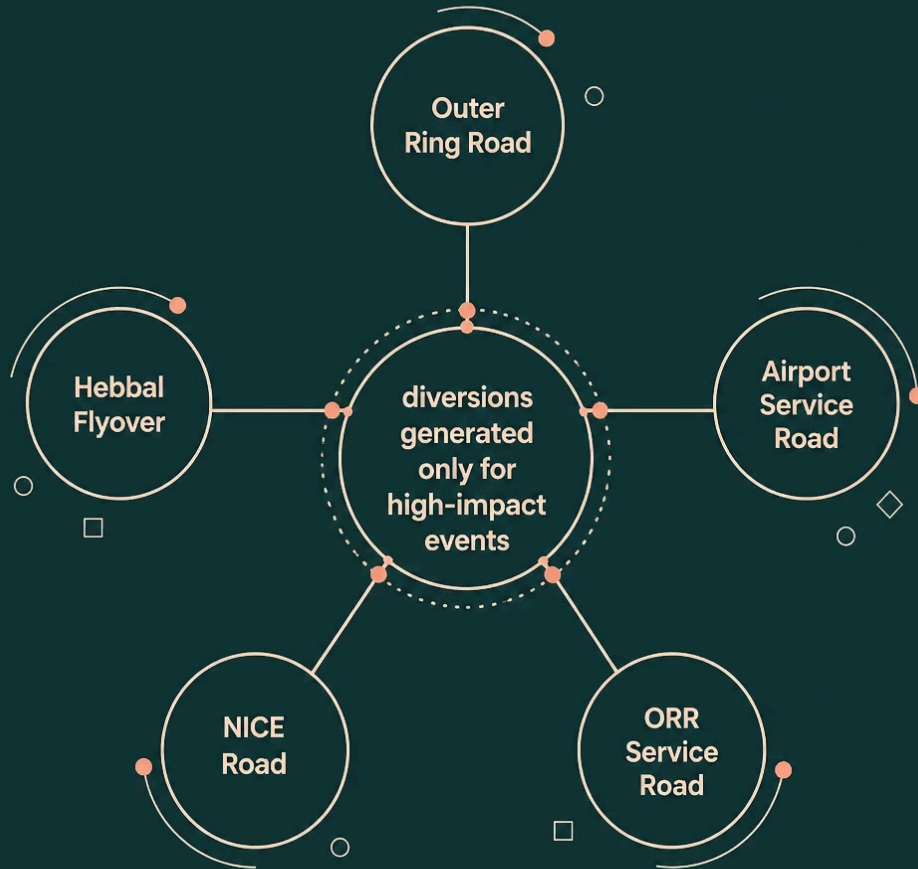


What EventPulse Recommends

- **Police deployment** — calibrated by event type and severity
- **Barricade count** — matched to road closure requirements
- **Dynamic scaling** — adjusts for peak hours and day-of-week patterns

i VIP movements demand the highest staffing at **34 officers** — reflecting their near-certain road closure rate.

Dynamic Routing & Mitigation Impact



Smart Diversion Logic

Route recommendations are generated **only for high-impact events**, drawing from Bengaluru's key arterial corridors.

Estimated Traffic Reduction

- Tree Fall – **55%** impact reduction
- VIP Movement – **50%** impact reduction
- Public Event – **43%** impact reduction

Post-Event Learning & Live Dashboard

Event Category	Avg Impact	Post - Mitigation	Officers	Learning Gap
VIP Movement	High	50%	34	High
Public Event	Medium	43%	22	Medium
Tree Fall	Medium	55%	20	Low
Breakdown	Low	60%	5	Low

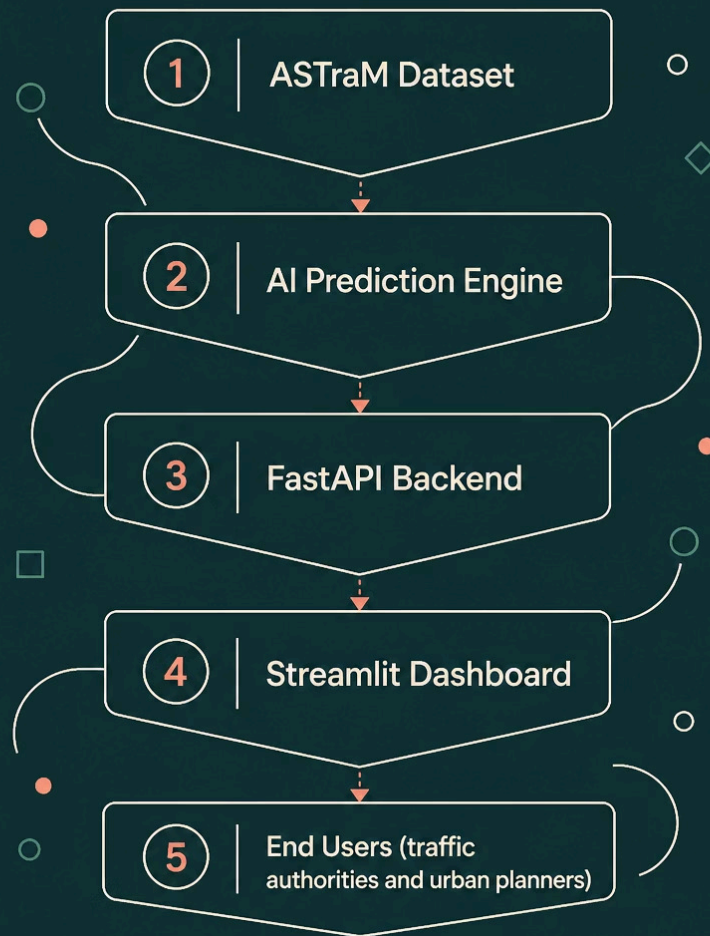
Learning Insights

VIP movements show the **highest mitigation opportunity** – making them the top priority for proactive planning.

AI Decision Dashboard

- Traffic hotspot & impact heatmaps
- Police requirement overlays
- Mitigation effectiveness analytics
- Learning gap leaderboard

Deployment & The Road Ahead



Live Deployment

- **Frontend:** Streamlit Dashboard
- **Backend:** FastAPI on Render
- **Stack:** Python · Pandas · FastAPI · Streamlit · Plotly · Folium

Future Roadmap

- Real-time CCTV integration
- Weather forecasting & Google Maps API
- Reinforcement learning for diversion optimization
- Adaptive signal timing & live emergency response

"Transforming historical event data into actionable intelligence for safer and smarter cities."